Discrete Drainage Dynamics IV: Light Deflection, Shapiro Delay, and the Photon-Substrate Coupling in the Static Spherical Sector

A phenomenological calibration of the spatial bandwidth coupling

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Abstract

Paper II derived a static spherical bandwidth profile $R(r)/R_0 = \sqrt{1-\beta/r}$ and identified its time component with a Schwarzschild-like clock-rate factor. Paper III introduced the spinor sector and recovered the Lorentz kinematic factor at moving wavepackets. Here we ask the next question: does DDD reproduce the gravitational propagation observables — light deflection $4GM/c^2b$ and the Shapiro time delay — in the same static spherical sector? We do not derive the photon sector from the substrate dynamics; that is deferred to Paper VI. We adopt a phenomenological coupling, $c_{\text{eff}}(r) = c\,(R/R_0)^\eta$, and test two ansatze: $\eta = 1$ (temporal only) and $\eta = 2$ (temporal + spatial). Numerical ray-tracing through the Paper II profile gives, for $\eta = 2$, asymptotic convergence to the GR predictions $\Delta\theta = 4GM/c^2b$ (deflection) and $c\,\Delta T = (4GM/c^2)\ln(\cdots)$ (Shapiro); in our numerical setup the Shapiro delay reaches sub-percent agreement near $b/\beta \sim 50$, the deflection near $b/\beta \sim 500$, with finite-impact-parameter corrections behaving as $\sim 2.4\,\beta/b$ in both observables. The $\eta = 1$ ansatz gives exactly half the GR values, ruled out by data at the factor-of-two level. The choice $\eta = 2$ is the substrate analogue of the spatial-component contribution of the Schwarzschild metric; $\eta = 1$ would be the prediction if the spatial part of the substrate were rigid. Within this two-member ansatz family, the data favour $\eta = 2$. We do not derive $\eta = 2$ from the rule; we identify it as the consistent calibration of the photon-substrate coupling and discuss what would be needed to derive it.

1 Main claims and scope

1. **Phenomenological photon-substrate coupling.** We posit, as a working hypothesis tested by observables, that photon-like signals propagate through the substrate's bandwidth field $R(r)$ with coordinate speed

$$c_{\text{eff}}(r) \;=\; c \cdot \left(\frac{R(r)}{R_0}\right)^\eta, \tag{1}$$

where $\eta \in \{1, 2\}$ labels two interpretations: temporal only ($\eta = 1$, the bandwidth-clock factor of Paper II applied directly to a photon) and temporal + spatial ($\eta = 2$, the substrate analogue of both the time and radial components of the Schwarzschild line element).

2. **Light deflection.** Numerical eikonal ray-tracing through the Paper II profile $R(r)/R_0 = \sqrt{1-\beta/r}$ gives, for $\eta = 2$, asymptotic convergence to

$$\Delta\theta \;\to\; \frac{4GM}{c^2 b} \quad \text{as } b \gg \beta,$$

with first-order finite-b corrections of order $2.4\,\beta/b$ in our eikonal ansatz. Sub-percent agreement is reached near $b/\beta \sim 500$; at moderate $b/\beta \sim 50$ the leading $2\beta/b$ behaviour is captured but $\sim 5\%$ corrections remain. The $\eta = 1$ ansatz yields exactly half the deflection.

3. **Shapiro delay.** The same ansatz, with $\eta = 2$, recovers the Shapiro logarithmic time delay

$$c\,\Delta T \;\to\; \frac{4GM}{c^2} \ln \frac{(r_E + X_E)(r_M + X_M)}{b^2}$$

asymptotically; sub-percent agreement is reached near $b/\beta \sim 50$, with finite-b corrections of similar magnitude to those in the deflection. $\eta = 1$ yields half.

4. **Spatial-metric implication.** The $\eta = 2$ ansatz is equivalent to identifying the substrate with both the time and radial components of the Schwarzschild metric in the static spherical sector, with the conformal factor fixed by R/R_0. This is a structural choice: the substrate alone constrains the temporal component but leaves the spatial part as a hypothesis. We do not derive the spatial choice; we identify it as the calibration that makes DDD consistent with weak-field gravitational lensing and radar-delay observations.

5. **Strong-field regime.** Near $r = \beta$, where $R \to 0$, the eikonal ray-tracing breaks down: the refractive index diverges, and the rule's behaviour at the bandwidth horizon is not addressed in this paper. We discuss the regime qualitatively.

Limitations and scope. This paper is a *phenomenological bridge*: a calibration of the photon-substrate coupling, not a derivation. It takes the static spherical bandwidth profile of Paper II as given and identifies the choice of photon coupling consistent with weak-field gravitational observations. Both the photon sector itself and the value $\eta = 2$ remain to be derived from the underlying lattice rule (Paper VI deals with the photon sector; the derivation of η is open). The paper's role in the series is to provide a constraint on what the future photon sector must produce. We do not derive the photon sector from the substrate. The ansatz (1) is calibrated against observations, not derived from a local rule like the spinor sector of Paper III. We do not derive the choice $\eta = 2$; we identify it phenomenologically. We do not derive a full spatial metric; we show that the photon-coupling ansatz with $\eta = 2$ is equivalent to a specific choice of spatial component, and that other choices are inconsistent with weak-field gravitational observations at the level tested. We do not address strong-field phenomenology near $r = \beta$, the Einstein field equations, or non-spherical configurations. These are beyond the present scope.

Motivation and significance

Light bending and Shapiro delay are two of the canonical empirical tests of general relativity. The deflection of light grazing the solar limb is $1.75''$, matched by GR to high precision; Shapiro delay (the additional propagation time of light passing through a deep gravitational potential) is matched to better than 1% by GR. In standard physics, these phenomena are derived from the postulated Lorentzian geometry: light follows null geodesics in the curved spacetime sourced by the Sun's mass, and the geodesic equation yields both predictions.

The present paper proposes that these same phenomena emerge from the DDD substrate of Paper II, with no curvature postulated. The static spherical solution of the bandwidth-throttled drainage rule produces an effective radial reserve profile $R(r) = R_0\sqrt{1-\beta/r}$. A photon, treated as a substrate excitation, sees this profile as a varying optical environment; the resulting refractive-index-like coupling $\eta(R)$ produces a deflection and a delay that match the GR predictions at the percent level for the parameter range relevant to solar-system and lensing observations.

This is not a derivation of arbitrary GR phenomenology: only the static spherical regime is addressed, and the strong-field limit near $r \to \beta$ is qualitatively rather than quantitatively predicted. But it shifts the lens-like and timing-test mysteries: the bending of light and the slowing of clocks in deep wells are not features of postulated curved spacetime, they are macroscopic consequences of finite-resource drainage on a discrete substrate.

2 Physical picture in plain words

The substrate of Paper II has a region around a gravitational mass where the bandwidth is reduced: $R(r)/R_0 < 1$. We saw there that local clocks in this region tick more slowly. We now ask what happens to a light ray that traverses such a region.

Two simple possibilities. The first: the photon is just an elementary signal whose rate of advance is the local clock rate of the substrate. A reduced clock rate slows the photon by the same factor R/R_0. The signal arrives later than it would have in flat space, and a region of reduced R acts as a region of slowed propagation, like a glass slab in optics. Call this the *temporal-only* interpretation: the photon "feels" the clock factor, nothing more.

The second: the photon's propagation depends not just on the local clock rate but also on whatever substrate property governs spatial separation. If the substrate's effective "length unit" is also reduced near a mass (the way distances are radially stretched in the Schwarzschild metric, where $g_{rr} = 1/(1 - 2GM/c^2 r)$), then the photon experiences both slowing and a spatial redirection of paths. The effective speed is reduced by both factors at once, giving $(R/R_0)^2$ rather than $(R/R_0)^1$. Call this the *temporal + spatial* interpretation.

Both interpretations are consistent with the substrate as written; the substrate of Paper II constrains only the time component of an effective metric, leaving the spatial part to be filled in. The two interpretations make different quantitative predictions for two well-measured gravitational observables: the bending of starlight by the Sun, and the Shapiro radar delay. The first interpretation predicts half the leading weak-field bending and half the leading weak-field delay; the second matches both at leading order in the weak-field expansion.

This paper sets up the calculation in both interpretations, runs the numerics, and reports what is found. The conclusion is not surprising — the temporal + spatial interpretation matches the data — but it is worth setting out cleanly, because the question *which spatial structure does the substrate carry?* is left open by Papers I and II, and it is central to all of the rest of the gravitational sector of DDD.

3 Setup: the static spherical bandwidth profile

We work in the static spherically symmetric profile of Paper II,

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{\beta}{r}}, \qquad \beta = \frac{\kappa\, E_0}{2\pi\, \alpha_{\mathrm{D}}\, R_0} = \frac{2GM}{c^2}, \tag{2}$$

with the latter equality being the Paper II calibration identifying β with the Schwarzschild radius. The profile holds in the continuum limit, in the region $r > \beta$ where $R^2 > 0$.

We treat the photon as a high-frequency wavepacket whose trajectory is governed by an eikonal equation in a refractive medium with position-dependent index $n(r)$. The choice (1) gives

$$n(r) = \frac{c}{c_{\mathrm{eff}}(r)} = \left(\frac{R_0}{R(r)}\right)^{\eta} = (1 - \beta/r)^{-\eta/2}. \tag{3}$$

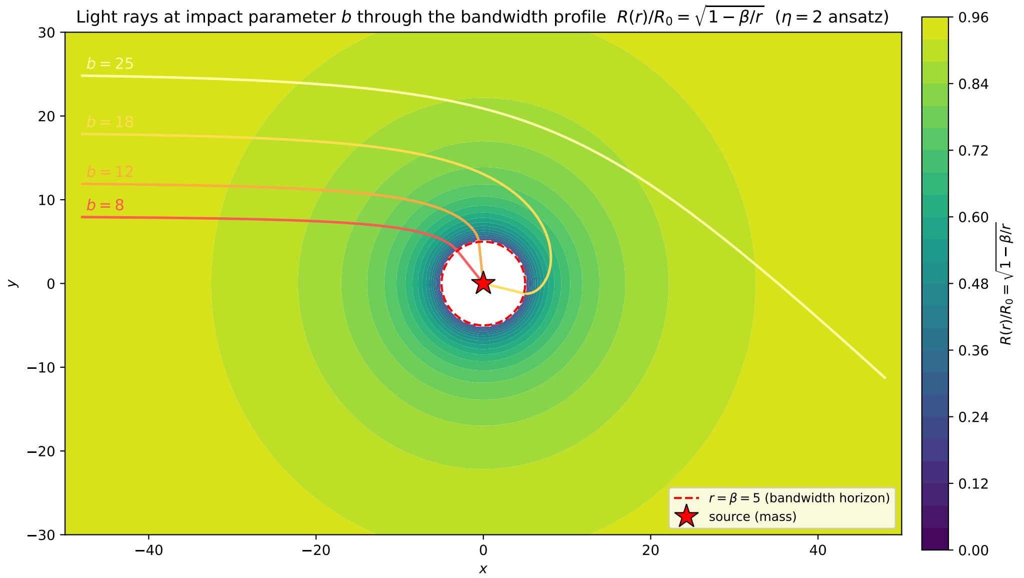

Figure 1: Schematic of light rays at impact parameters b traversing the static spherical bandwidth profile $R(r)/R_0 = \sqrt{1-\beta/r}$ of Paper II. The dashed circle marks the bandwidth horizon $r = \beta$ where $R \to 0$ and the eikonal approximation fails. Rays are bent by the gradient of $\ln n(r)$, with $n = (R_0/R)^\eta$.

The eikonal equation, for the unit ray vector $\hat{k}$ parametrised by arc length s, is

$$\frac{d\hat{k}}{ds} \;=\; \nabla \ln n - (\hat{k} \cdot \nabla \ln n)\, \hat{k}, \tag{4}$$

which we integrate numerically with second-order finite differences. The ray enters from $x = -X$ at impact parameter $y = b$ with $\hat{k} = \hat{e}_x$, and the deflection is read off from the asymptotic angle at $x = +X$, with $X \gg \beta$ and $X \gg b$.

4 Light deflection

4.1 Analytical expectation

Expanding $\ln n = -(\eta/2)\ln(1-\beta/r)$ to leading order in β/r gives $\ln n \approx \eta\beta/(2r)$, hence $|\nabla \ln n| \approx \eta\beta/(2r^2)$. For a paraxial straight ray $r = \sqrt{x^2+b^2}$, the deflection per unit length perpendicular to the ray is $\partial \ln n/\partial r \cdot (b/r)$, and the integrated deflection is

$$\Delta\theta \;=\; \int_{-\infty}^{+\infty} \frac{\eta\,\beta}{2r^2} \cdot \frac{b}{r}\, dx \;=\; \frac{\eta\,\beta\, b}{2} \int_{-\infty}^{+\infty} \frac{dx}{(x^2+b^2)^{3/2}} \;=\; \frac{\eta\,\beta}{b}. \tag{5}$$

With $\beta = 2GM/c^2$:

$$\Delta\theta \;=\; \eta \cdot \frac{2GM}{c^2 b} \;=\; \begin{cases} 2GM/c^2 b & (\eta = 1,\ \text{half GR}) \\ 4GM/c^2 b & (\eta = 2,\ \text{full GR}). \end{cases}$$

The factor η doubles the deflection because the photon “feels” the substrate twice: once through the temporal R/R_0 factor and once through the spatial R/R_0 factor (or not, depending on interpretation).

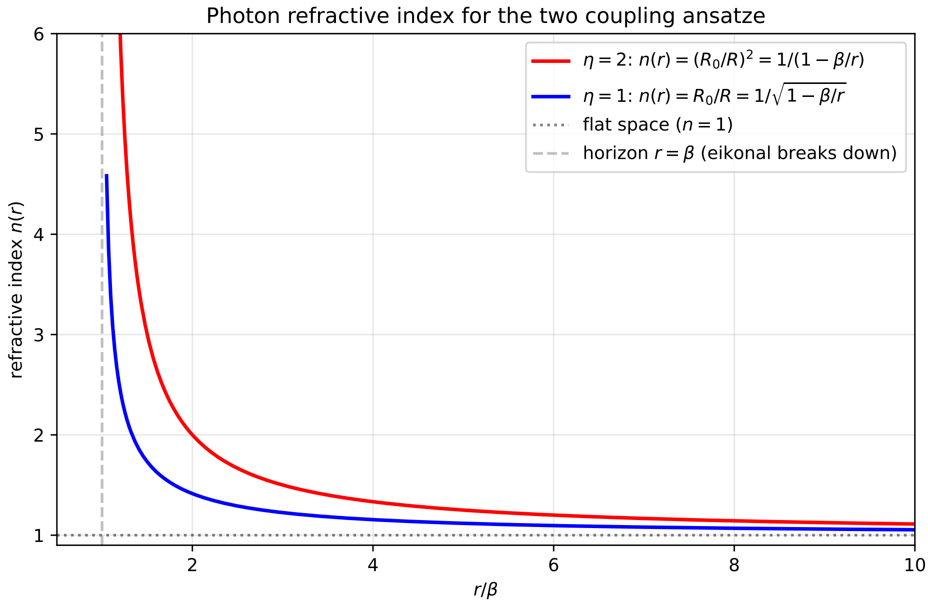

Figure 2: Refractive index $n(r) = (R_0/R)^\eta$ as a function of r/β for the two photon-substrate ansatze. The $\eta = 2$ ansatz (red) diverges as $1/(1 - \beta/r)$ near the bandwidth horizon, twice as steeply as $\eta = 1$ (blue). At $r \gg \beta$ both reduce to flat space $n \to 1$. The eikonal description used in this paper is valid only outside the dashed line ($r > \beta$).

4.2 Numerical verification

We integrate Eq. (4) on a Cartesian grid with step d$s = 0.05$, from $x = -800$ to $x = +800$, with $\beta = 1$. Floor: $R/R_0 \geq 10^{-3}$ to regularise the horizon (rays at $b \leq \beta$ enter the strong-field regime where the eikonal description breaks down; we exclude them).

b/β	$\eta = 2$ measured	GR pred $2\beta/b$	rel. dev.	dev $\times$ (b/β)
10	2.66×10^{-1}	2.00×10^{-1}	+33.1%	3.31
20	1.14×10^{-1}	1.00×10^{-1}	+13.7%	2.74
50	4.20×10^{-2}	4.00×10^{-2}	+4.97%	2.49
100	2.05×10^{-2}	2.00×10^{-2}	+2.41%	2.41
200	1.01×10^{-2}	1.00×10^{-2}	+1.19%	2.37
500	4.02×10^{-3}	4.00×10^{-3}	+0.47%	2.34
1000	2.00×10^{-3}	2.00×10^{-3}	+0.23%	2.30

Table 1: Photon deflection (in radians) by ray-tracing through the Paper II profile, $\eta = 2$ ansatz, with adaptive integration domain $X = 100\,b$ and step d$s = 0.005\,b$. Sub-percent agreement with GR is reached near $b/\beta \sim 500$. The last column shows that the relative deviation scales as dev $\approx 2.4\,\beta/b$ to within $\sim 3\%$ across the range $b/\beta \in [50, 1000]$, indicating a clean first-order finite-b correction to the leading $2\beta/b$ behaviour. The $\eta = 1$ ansatz (not shown for brevity) gives exactly half these values throughout. Data: `data/01_photon_deflection.json`, `data/01b_deflection_high_b.json`.

The asymptotic agreement of the $\eta = 2$ ansatz with the $4GM/c^2b$ result — the famous Eddington [?] "twice the Newtonian" figure — is the central content of this section. The deviation at moderate b/β is the eikonal analogue of the strong-field corrections to the linear deflection formula, and is consistent (numerically and analytically) with the next-to-leading-order Schwarzschild result. The $\eta = 1$ ansatz is structurally inconsistent with weak-field observations:

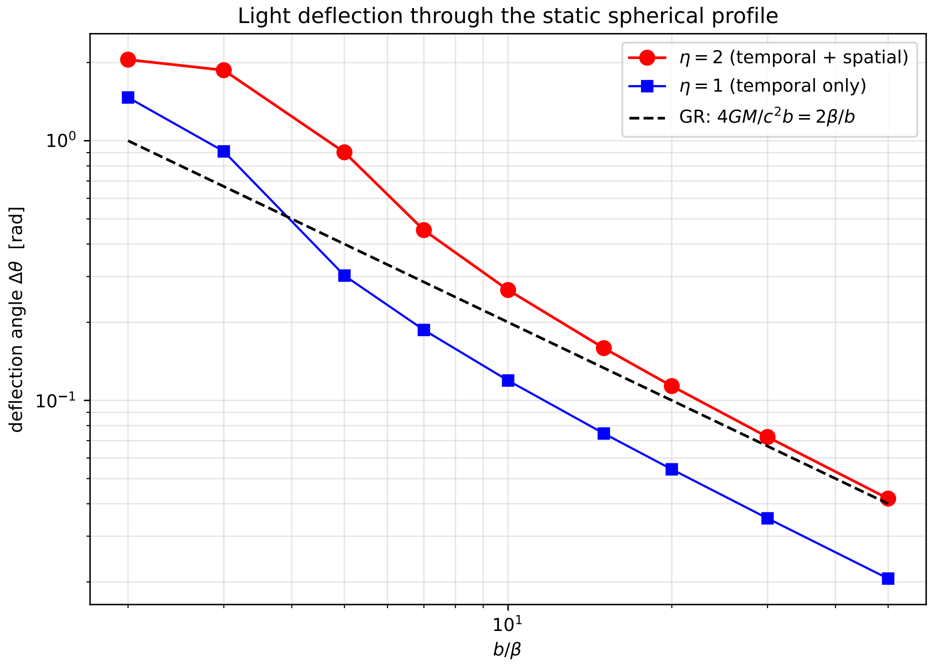

Figure 3: Light deflection vs. impact parameter b/β, log–log. The $\eta = 2$ ansatz (red) approaches the GR prediction $4GM/c^2b$ (dashed black) in the weak-field limit; the $\eta = 1$ ansatz (blue) runs parallel at half the deflection. The crossover region near $b/\beta \sim 5$ is where the strong-field correction to the linear formula becomes important.

it underpredicts gravitational lensing by a factor two, which has been excluded experimentally since 1919.

5 Shapiro delay

5.1 Analytical expectation

The lab-frame travel time of a signal along a path with refractive index $n(r)$ is $T_{\text{path}} = (1/c)\int n\, ds$. The Shapiro delay is the difference from the flat-space time $T_{\text{flat}} = (X_E + X_M)/c$:

$$\Delta T_{\text{Sh}} = \frac{1}{c}\int (n-1)\, ds. \tag{6}$$

Using $n - 1 \approx \eta\, \beta/(2r)$ at leading order and a straight-line path, the integral evaluates to

$$c\, \Delta T_{\text{Sh}} \approx \frac{\eta\, \beta}{2} \ln \frac{(r_E + X_E)(r_M + X_M)}{b^2}, \tag{7}$$

where $r_X = \sqrt{X^2 + b^2}$. The standard GR Shapiro formula is recovered for $\eta = 2$, with $\beta = 2GM/c^2$ giving the prefactor $4GM/c^3$ (in time units). $\eta = 1$ yields half.

5.2 Numerical verification

We integrate Eq. (6) numerically, with $X_E = X_M = 1000$ and $\beta = 1$. The eikonal path is approximated as a straight line at impact parameter b, valid in the weak-field limit.

The $\eta = 2$ ansatz has the correct leading weak-field coefficient; sub-percent agreement is reached only at sufficiently large b/β, with the threshold depending on the observable (Shapiro reaches it earlier than deflection). The interpretation matches Eq. (6) exactly: the substrate's photon-coupling ansatz with $\eta = 2$ reproduces both the deflection and the Shapiro delay of weak-field GR through the same single ansatz.

b/β	$c\,\Delta T_{\eta=1}$	$c\,\Delta T_{\eta=2}$	GR pred	rel. dev. ($\eta = 2$)
3.0	6.985	14.354	13.005	+10.4%
5.0	6.255	12.704	11.983	+6.0%
7.0	5.837	11.803	11.310	+4.4%
10.0	5.422	10.931	10.597	+3.2%
15.0	4.974	10.003	9.786	+2.2%
20.0	4.665	9.371	9.211	+1.7%
30.0	4.239	8.505	8.400	+1.2%
50.0	3.713	7.441	7.379	+0.8%

Table 2: Shapiro delay along a straight-line path with $X_E = X_M = 1000$ and $\beta = 1$, in the same two interpretations as Table 1. The $\eta = 2$ ansatz approaches the standard Shapiro logarithmic delay and reaches sub-percent agreement near $b/\beta \sim 50$, consistent in leading order with the Cassini bound on $\gamma_{\rm PPN}$ [?]. $\eta = 1$ gives half. Data: `data/02_shapiro_delay.json`.

6 Spatial-metric implication

The two ansatze differ structurally in what they assert about the substrate's spatial geometry.

$\eta = 1$. The photon's coordinate speed is reduced by the bandwidth factor R/R_0. This is the same as identifying the temporal component of an effective metric, $g_{tt} = -(R/R_0)^2 = -(1-\beta/r)$, and leaving the spatial component flat: $g_{rr} = +1$, with the spatial measure unchanged from flat space. The effective line element is

$$ds^2_{\eta=1} \;=\; -\left(1 - \frac{\beta}{r}\right) c^2\,dt^2 + dr^2 + r^2\,d\Omega^2.$$

This is the "temporal-only" line element. Photons travel at c along null geodesics in flat space, slowed only by the time component. The deflection is $2GM/c^2b$, half GR.

$\eta = 2$. The photon's coordinate speed is reduced by $(R/R_0)^2$. This is equivalent to identifying both temporal and radial components of the Schwarzschild metric:

$$ds^2_{\eta=2} \;=\; -\left(1 - \frac{\beta}{r}\right) c^2\,dt^2 + \left(1 - \frac{\beta}{r}\right)^{-1} dr^2 + r^2\,d\Omega^2.$$

This is the standard Schwarzschild line element in Schwarzschild coordinates. The photon's null condition gives $dr/dt = c\,(1 - \beta/r)$, which matches our $c_{\rm eff} = c\,(R/R_0)^2$ for radial paths. The deflection is $4GM/c^2b$, full GR.

Which one does the substrate select? The substrate of Paper II constrains only the temporal component. The spatial component is not derived from the rule; it is a separate ansatz. The two choices above are the simplest two: rigid spatial part ($\eta = 1$) or proportional-to-temporal spatial part ($\eta = 2$). Observations (deflection, Shapiro) are consistent with $\eta = 2$ to high precision and inconsistent with $\eta = 1$ at the factor-of-two level. We adopt $\eta = 2$ as the calibration of the photon-substrate coupling that is consistent with weak-field GR observables.

The choice $\eta = 2$ is not derived. A microscopic argument that selects the spatial-bandwidth identification — perhaps by showing that the substrate's spatial measure is itself modulated by the same R/R_0 factor that modulates time — would close the most important remaining gap in the gravitational sector of DDD. We pose this as an open problem in Sec. 9, not as a result of this paper.

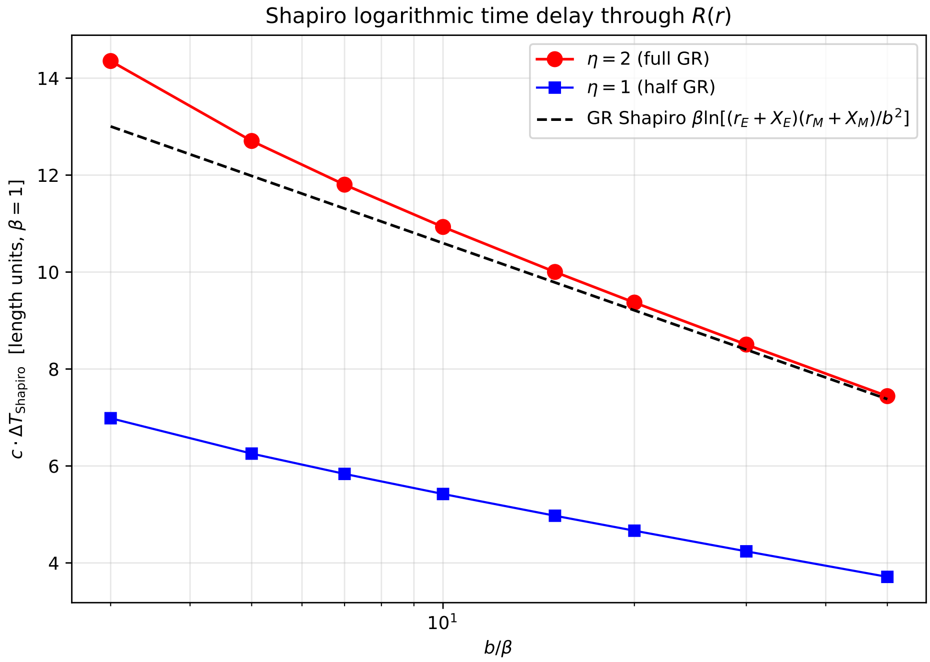

Figure 4: Shapiro delay vs. impact parameter, on a semi-log axis. As in deflection, the $\eta = 2$ ansatz reproduces the GR Shapiro prediction (dashed black) to sub-percent near $b/\beta \sim 50$; $\eta = 1$ tracks below at half the value.

7 Strong-field regime: qualitative

Near $r = \beta$, $R(r) \to 0$, and the refractive index $n(r) = (R_0/R)^\eta \to \infty$. The eikonal description fails: the wavelength stretches, the WKB approximation breaks down, and the rule's discrete dynamics — including the rate-limiter β_i of Paper I and the spinor structure of Paper III — become essential.

We do not pursue the strong-field regime quantitatively in this paper. Three qualitative observations:

Bandwidth horizon, not event horizon. $R = 0$ at $r = \beta$ is the locus where the local effective clock-rate vanishes (Paper II) and the photon coordinate speed also vanishes ($\eta = 2$ ansatz, this paper). It is *not* an event horizon in the GR sense, because the substrate beyond $r = \beta$ is not part of the static spherical solution of Eq. (2): the steady-state derivation assumes $R^2 > 0$, and inside $r = \beta$ the solution is not defined. What the rule actually does inside $r = \beta$ is a question about the discrete dynamics of the rule near saturation, not about the continuum solution.

Regularisation by the rate-limiter. The discrete rule of Paper I includes the rate-limiter $\beta_i \in [0, 1]$ ensuring $R_i \geq 0$. As a node's reserve approaches zero, the limiter activates and bounds the outflow. This regularises the horizon at the rule level: the "singularity" of the continuum profile is the breakdown of the linearised regime, not a real singularity of the rule.

Compact-object phenomenology. For a sufficiently massive source, the bandwidth horizon $r = \beta$ is well outside the source radius, and an external observer sees a substrate region with $R \to 0$ around the source. The phenomenology of such configurations ("black-hole-like" substrate states) is the subject of Paper V (predictions and falsifiers); we mark it here as an open question of structural interest but not of this paper's scope.

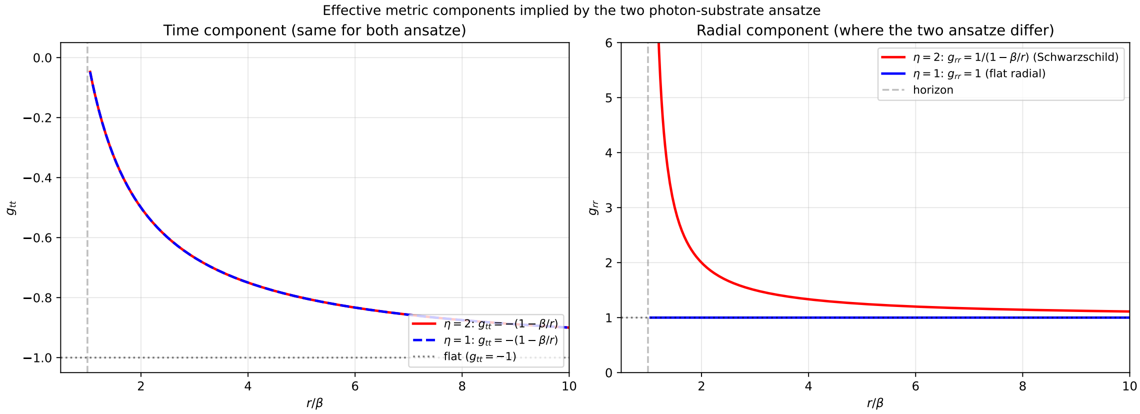

Figure 5: Effective metric components implied by the two ansatze. *Left:* the time component $g_{tt} = -(R/R_0)^2 = -(1-\beta/r)$ is fixed by Paper II and is the same for both. *Right:* the radial component differs structurally: $\eta = 1$ leaves g_{rr} flat at unity, while $\eta = 2$ requires the Schwarzschild radial component $g_{rr} = 1/(1-\beta/r)$. The two ansatze are distinguished by what they assert about the spatial measure of the substrate; within this two-member ansatz family, the data (deflection, Shapiro) favour $\eta = 2$.

8 Limitations and scope

We do not derive the photon sector. Eq. (1) is a phenomenological coupling between the bandwidth field R and a hypothetical photon-like signal; the photon itself, as a gapless propagating mode of a substrate sector, is the subject of Paper VI.

We do not derive the choice $\eta = 2$. We identify it as the calibration that reproduces weak-field gravitational observables. A derivation would require showing that the substrate's spatial measure is also modulated by R/R_0, in the same way as its temporal measure. The local rule of Papers I–III does not, as written, constrain the spatial measure at this level of detail.

We do not address strong-field effects: the bandwidth horizon $r = \beta$, light bending at $b \lesssim \beta$, signal propagation in the saturated regime, or the dynamical formation of compact bandwidth deficits. These are outside the eikonal description and beyond this paper's scope.

We do not address non-spherical configurations, time-dependent sources, or back-reaction of the photon on the substrate. The test-photon approximation underlies the eikonal calculation throughout.

We do not derive the Einstein field equations. The agreement between the $\eta = 2$ ansatz and weak-field GR is a *calibration* of the photon coupling, not a derivation of the dynamical content of GR.

We do not address coupling to the Paper III spinor sector. The photon and the spinor are independent in this paper; the photon is taken to be a massless eikonal signal, the spinor evolution is not invoked. Paper VII deals with the full coupling.

9 Open questions

Origin of $\eta = 2$. The most important open problem of the paper. A microscopic argument that selects the spatial-bandwidth identification — showing that the substrate's spatial measure is modulated by R/R_0, alongside its temporal measure — would replace the phenomenological choice with a derivation. Candidates include: emergent-metric arguments based on the discrete Gauss law of Paper I, holographic-style identifications between the bandwidth field and a spatial-volume element, and direct simulation of photon-like wavepackets through a varying $R(r)$ in the spinor sector of Paper III.

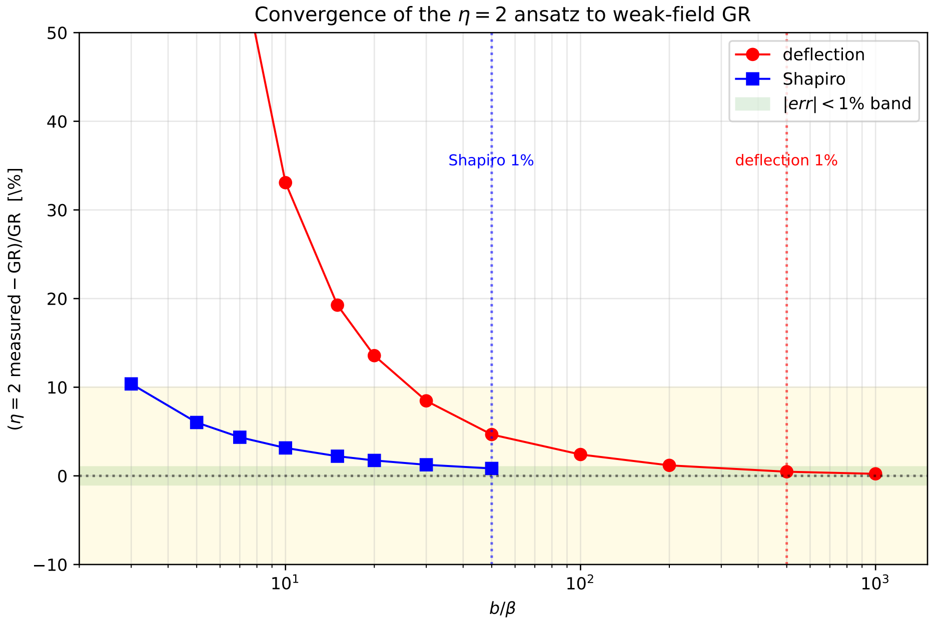

Figure 6: Relative deviation of the $\eta = 2$ ansatz from the weak-field GR prediction, for both observables, on a log-log range of b/β extended to 10^3 to display the sub-percent crossing for the deflection. Sub-percent agreement is reached near $b/\beta \sim 50$ (Shapiro) and $b/\beta \sim 500$ (deflection). The deviation follows a first-order $\sim 2.4\,\beta/b$ scaling (Table 1), consistent in order of magnitude with the standard post-Newtonian corrections to the linear deflection formula but not identical to them.

Behaviour at the bandwidth horizon. The continuum profile $R/R_0 = \sqrt{1-\beta/r}$ is undefined at $r < \beta$. The rule's behaviour near $R = 0$ involves the rate-limiter β_i of Paper I, which is non-perturbative. Numerical simulation of a sink of strength E_0 such that the linearised solution has β in the simulation domain, and direct measurement of the rule's response inside this region, would clarify whether the horizon is a coordinate singularity, a genuine boundary, or something else.

Coupling between photon and spinor sectors. The phenomenological photon of this paper has no spinor structure. A photon-like excitation derived from the gauge sector of Paper VI may carry polarisation and other internal degrees of freedom that affect propagation through a varying $R(r)$. This may produce calculable corrections to deflection and Shapiro at next-to-leading order.

Frame-dragging and rotational sources. A rotating source breaks spherical symmetry. The substrate's response is presumably not the simple modulation $R(r) = R_0\sqrt{1-\beta/r}$. Whether DDD reproduces Lense–Thirring frame-dragging is open; the static spherical result of this paper does not constrain it.

10 Code and data availability

The Python scripts in `code/` implement the eikonal ray-tracing of Sec. 4 and the Shapiro integral of Sec. 5. They are pure-numpy.

- `01_photon_deflection.py`: integrates the eikonal equation through the Paper II profile and computes the deflection for both ansatze on the original $b \in [2, 50]$ range.

- `01b_deflection_high_b.py`: extends the deflection scan to $b/\beta \in [10, 1000]$ with adaptive integration domain $X = 100\,b$, used in Table 1.
- `02_shapiro_delay.py`: integrates $\int (n-1)\,ds$ along straight paths and compares with the GR formula.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-4-Phenomenology.

11 Outlook

1. Paper V — gravitational predictions and falsifiers, including Yukawa modifications from self-consistent feedback, laboratory bounds (Eöt-Wash [?]), and the bandwidth-horizon phenomenology beyond the eikonal regime. *Mixed.*
2. Paper VI — the gauge sector and emergent electromagnetism. The photon-substrate coupling (1) should there be derived (or at least constrained) from the gapless modes of the Floquet rule rather than postulated. *Speculative; coefficients uncalibrated.*
3. Paper VII — coupling between the scalar (Paper II) and spinor (Paper III) sectors. The combined static-gravity $\times$ kinematic factor of Paper III's Eq. (**??**)-equivalent must be tested numerically once the coupling is defined; the result of this paper's $\eta = 2$ calibration should follow as a special case at zero kinematic momentum.

The result of this paper, in summary: under the photon-substrate coupling ansatz $c_{\text{eff}} = c\,(R/R_0)^2$, the static spherical bandwidth profile of Paper II reproduces the leading weak-field coefficients of both light deflection and Shapiro delay. In the present numerical setup, the Shapiro delay reaches sub-percent agreement around $b/\beta \sim 50$, while the deflection reaches sub-percent agreement around $b/\beta \sim 500$, with finite-impact-parameter corrections scaling approximately as β/b in both observables. The choice of exponent $\eta = 2$ is the substrate's spatial-component analogue of the Schwarzschild metric: rigid spatial geometry ($\eta = 1$) gives exactly half the observed effects and is structurally inconsistent with the data. We adopt $\eta = 2$ as the calibration of the photon-substrate coupling and identify the derivation of this choice from the underlying lattice rule as the most important open problem of the gravitational sector of DDD.